Global Sorting, Morse Indices, and the Route-A Boundary of the Brockett Double-Bracket Flow

Route-A structural certificate C185

Abstract

For every real symmetric simple-spectrum orthogonal orbit and every strictly increasing diagonal target, we close the Brockett flow $\dot{H} = [H, [H, N]]$ without a dimension cutoff. The flow is globally isospectral and satisfies $d\,\mathrm{Tr}(HN)/dt = \|[H, N]\|_F^2$. Its $n!$ equilibria are diagonal permutations. Each tangent pair has an explicit rate whose positive signs are precisely the inversions; hence the Morse index of the sorting energy is the inversion number. Every trajectory converges, generic data sort, and no nonconstant trajectory is recurrent. Repeated spectra are isolated as a degenerate boundary; target repetition supplies its Morse–Bott component. Despite this complete dynamical theorem, arbitrary real inputs give no arithmetic origin or primitive periodic carrier, so Route A is rejected.

Keywords: double bracket; isospectral flow; matrix sorting; Morse index; gradient dynamics.

中文摘要

本文对任意维实对称单谱正交轨道与严格递增对角目标，完整刻画 $\dot{H} = [H, [H, N]]$：解全局存在并保持谱，势函数导数恰为交换子的 Frobenius范数平方；全部平衡点是n!个对角排列。 每个切向二元模态的符号由排列逆序决定，从而排序能量的 Morse指标等于逆序数； 所有轨道收敛，几乎所有初值完成排序，且不存在非常值回复或周期轨道。 重谱另属退化边界，其中目标重谱产生Morse–Bott临界族。 该系统没有内生素数来源或原始周期载体，故路线A终止。

关键词：双括号流；等谱流；矩阵排序； Morse指标；梯度动力学。

1 Frozen family and global identities

Let

$$\lambda_1 < \cdots < \lambda_n, \qquad \nu_1 < \cdots < \nu_n,$$

put $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$ and $N = \mathrm{diag}(\nu_1, \ldots, \nu_n)$, and work on $\mathcal{O}_\Lambda = \{Q\Lambda Q^\mathsf{T} : Q \in O(n)\}$. The equation and its gradient sorting role are classical Brockett [1]; no priority is claimed here.

Set $K(H) = [H, N]$. It is skew-symmetric and $\dot{H} = [H, K(H)]$. If $\dot{Q} = -K(H)Q$, orthogonality is preserved and $H = Q\Lambda Q^\mathsf{T}$ solves the equation. Thus the compact orbit is invariant, all spectral traces are constant, and the smooth vector field is complete. Cyclicity of trace gives the strict identity

$$\frac{d}{dt}\,\mathrm{Tr}(HN) = \mathrm{Tr}([H, K]N) = -\,\mathrm{Tr}(K^2) = \|K\|_F^2. \tag{1}$$

Consequently equilibrium is equivalent to $[H, N] = 0$. The strict target makes H diagonal, and the simple orbit spectrum makes its diagonal a permutation of the λ_i. There are exactly $n!$ equilibria $D_\pi = \mathrm{diag}(\lambda_{\pi(1)}, \ldots, \lambda_{\pi(n)})$.

Structure	Exact consequence	Boundary
skew Lax generator	global isospectral flow	not a fixed quantum operator
squared-norm derivative	strict gradient ordering	no nonconstant recurrence
simple/strict spectra	$n!$ isolated equilibria	target repetitions give zero modes

2 Pair modes, Morse index, and convergence

At $D = D_\pi$, write $H = D + \varepsilon X$. Since $[D, N] = 0$, linearization gives

$$L_D X = [D, [X, N]], \qquad (L_D X)_{ij} = (\lambda_{\pi(i)} - \lambda_{\pi(j)})(\nu_j - \nu_i)X_{ij}. \tag{2}$$

The tangent space consists of arbitrary symmetric off-diagonal pair modes. For $i < j$, the second factor in (2) is positive, while the first is positive exactly when $\pi(i) > \pi(j)$. Hence the ascent flow has $\mathrm{inv}(\pi)$ unstable modes and $\binom{n}{2} - \mathrm{inv}(\pi)$ stable modes. Equivalently, the Morse index of $-\,\mathrm{Tr}(HN)$ is $\mathrm{inv}(\pi)$. This sign convention is essential: it is not the Morse index of the increasing height.

The rearrangement inequality makes increasing alignment the unique height maximum and the only stable equilibrium. The bounded height has a limit; LaSalle invariance places every omega-limit set among the finite equilibria. Connectedness of an omega-limit set then forces convergence to a single one. All equilibria are hyperbolic by (2). Stable manifolds of nonsorted equilibria have lower dimension, so their finite union has zero orbit volume; every other initial point converges to the sorted diagonal. Finally, (1) is strict on a nonconstant trajectory. A recurrent return would restore an old height value, which is impossible; periodicity is a special case.

3 Degenerate boundary and Route-A stop

Repeated source eigenvalues collapse distinct diagonal permutations. A zero rate in (2) for an equal-eigenvalue pair is then a stabilizer/non-tangent direction on the lower-dimensional orbit, not a tangent zero mode. If N repeats, rotations inside a repeated N-eigenspace yield genuine tangent zero modes and continuous non-diagonal commuting families. This is the Morse–Bott component of a separate repeated-spectrum boundary. We claim neither its full classification nor a Bruhat/Schubert cell-closure theorem; neither is needed for the generic simple-spectrum result.

The exact artifact exhausts 5,912 permutation equilibria and 118,004 pair modes for $2 \le n \le 7$, plus six rational Lax/Lyapunov sentinels. An implementation-independent checker passes 183,158 assertions; a separate SymPy path passes 253,765 checks; replay is byte exact; 68 hostile mutations are rejected. These finite rows test the implementation, while the argument above proves the theorem for all n.

Route A stops at its entry gate. The two spectra are arbitrary ordered real data, not an intrinsic rational-prime source, and continuous time is not a logarithmic prime clock. Equation (1) removes every nonconstant primitive periodic carrier. A tangent characteristic polynomial is not a global dynamical determinant, and no target divisor, functional equation, counting law, continuation, or Weil compression is present. Orthogonal conjugation is generated by the state-dependent $K(H)$; it gives only a formal A4 hint, not a fixed linear quantum operator. Thus

$$(A0, A1, A2, A3, A4) = (\mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FAIL}, \mathrm{FORMAL\ HINT}),$$

overall Route A is rejected and Route B is false.

Revision-round focus. Round 2 locks the repeated-spectrum boundary, exact evidence populations, classical attribution, and the state-dependent-generator Route-A limitation.

Limitations and nonclaims. The paper does not claim novelty for the Brockett flow, a full degenerate critical-set or cell-closure theorem, arithmetic local data, an Euler factor, a root number, automorphy, a target divisor, a Hilbert–Polya operator, or Route-B authorization. The single source is an ownership lock, not an exhaustive literature review. Internal drafting checks are not external peer review.

Declarations

Data and code availability. Package-local exact evidence and deterministic code accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Funding. No external funding is reported.

Conflicts of interest. None known.

AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

References

[1] R. W. Brockett, "Dynamical systems that sort lists, diagonalize matrices, and solve linear programming problems," *Linear Algebra and its Applications* 146 (1991), 79–91. DOI: 10.1016/0024-3795(91)90021-N.